

prevent issues, let's turn off the Neutron server on the network controller and Neutron's Open vSwitch agents on all hosts.

Step 2: Configuring Open vSwitches in the controller and compute nodes

The Neutron plugin in every node must be removed because only OpenDaylight will be controlling the Open vSwitches. So, on each host, we will erase the pre-existing Open vSwitch config and set OpenDaylight to manage the Open vSwitch:

```
$ apt-get purge neutron-plugin-openvswitch-agent
$ service openvswitch-switch stop
$ rm -rf /var/log/openvswitch/*
$ rm -rf /etc/openvswitch/conf.db
$ service openvswitch-switch start
$ ovs-vsctl show
# The above command must return the empty set except
OpenVswitch ID and it's Version.
```

Step 3: Connecting Open vSwitch to OpenDaylight

Use the command given below to make OpenDaylight administer Open vSwitch:

```
$ ovs-vsctl set-manager tcp:<OPENDAYLIGHT MANAGEMENT IP>:6640
```

You can copy the Open vSwitch ID from the command `ovs-vsctl show`. Execute the above command in all the nodes (controller and compute nodes) to set ODL as the manager for Open vSwitch:

```
$ ovs-vsctl show
```

The above command will show that you are connected to the OpenDaylight server, which will automatically create a `br-int` bridge.

```
[root@vinoth ~]# ovs-vsctl show
9e3b34cb-fefc-4br4-828s-084b3e55rtfd
Manager "tcp:192.168.2.101:6640"
Is_connected: true
Bridge br-int
Controller "tcp:192.168.2.101:6633"
fail_mode: secure
Port br-int
Interface br-int
ovs_version: "2.1.3"
```

If you get any error messages during bridge creation, you may need to log out from the OpenDaylight Karaf console and check the `90-vtn-neutron.xml` file from the following path `distribution-karaf-0.5.0-Boron/etc/opendaylight/karaf/`.

The contents of `90-vtn-neutron.xml` should be as follows:

```
bridgename=br-int
```

```
portname=eth1
protocol=OpenFlow13
failmode=secure
```

By default, if `90-vtn-neutron.xml` is not created, VTN uses `ens33` as the port name.

After running the ODL controller, please ensure it listens to the following ports: 6633, 6653, 6640 and 8080.

Note:

- 6633/6653 are the OpenFlow ports.
- 6640 is the OVS Manager port.
- 8080 is the port for the REST API.

Step 4: Configure ml2_conf.ini for the ODL driver

Edit `vi /etc/neutron/plugins/ml2/ml2_conf.ini` in all the required nodes and modify the following configuration. Leave the other configurations as they are.

```
[ml2]
type_drivers = local
tenant_network_types = local
mechanism_drivers = opendaylight
[ml2_odl]
password = admin
username = admin
url = http://<OPENDAYLIGHT SERVER's IP>:8080/controller/nb/v2/neutron
```

Step 5: Configure the Neutron database

Reset the Neutron database, as follows:

```
$ mysql -uroot -p
$ drop database neutron;
$ create database neutron;
$ grant all privileges on neutron.* to 'neutron'@'localhost'
identified by '<YOUR NEUTRON PASSWORD>';
$ grant all privileges on neutron.* to 'neutron'@'%'
identified by '<YOUR NEUTRON PASSWORD>';
$ exit
$ su -s /bin/sh -c "neutron-db-manage --config-file /etc/neutron/neutron.conf --config-file /etc/neutron/plugins/ml2/ml2_conf.ini upgrade head" neutron
```

Restart the Neutron-server, as follows:

```
$ service neutron-server start
```

Step 6: Install the Python-networking-odl Python module

IMPORTANT: You should get the status alert if the Neutron service fails to start by this time. Don't worry. This is a temporary issue since you have enabled OpenDaylight as the `mechanism_driver` but not yet installed the Python module for it.

Install the `Python-networking-odl` Python module, as follows: